

Mohsin Furkh Dar

Assistant Professor (Research Faculty) · School of Computer Science, UPES, Dehradun, India

Deep Learning · Medical Image Analysis · Explainable & Uncertainty-Aware AI

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RESEARCH PROFILE

Computer scientist working at the intersection of **deep learning, medical image analysis, and soft computing**. My research designs architectures and learning objectives for medical image segmentation and classification, with an emphasis on *uncertainty-aware learning* (fuzzy rough set losses), *adaptive ensemble objectives*, and *interpretable attention mechanisms* for breast ultrasound diagnosis.

Nine peer-reviewed publications — seven as first author, four in **Q1 SCIE** journals — including *Computerized Medical Imaging and Graphics*, *Biomedical Signal Processing and Control*, *Image and Vision Computing*, *Ultrasound in Medicine & Biology* and *Neural Processing Letters*; UGC-NET JRF at All India Rank 53; a competitively awarded international travel grant; and reviewing for IEEE TMI, IEEE JBHI and Information Fusion.

Independent research programme. Since joining UPES I have established a line of work that is independent of my doctoral supervision: I am **Principal Investigator** on a ₹25 lakh ICMR ANVESHAN extramural proposal (under review) with a clinical Co-PI, and first and corresponding author on *MSCT-Trans* (Ultrasound in Medicine & Biology, 2026) — conceived, funded and executed at UPES with my own postgraduate student and without my doctoral advisor.

Forward agenda. Building clinically deployable, explainable AI for multimodal medical imaging — extending uncertainty-aware loss design and foundation-model adaptation from breast ultrasound to obstetric Doppler and other modalities, in partnership with clinical collaborators, and growing an externally funded research group in trustworthy medical AI.

9 PUBLICATIONS	7 FIRST AUTHOR	108+ CITATIONS	4 H-INDEX	4 Q1 SCIE PAPERS	8 JOURNALS RE- VIEWED FOR
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EDUCATION

Nov 2020 – Jul 2026	Ph.D. in Computer Science DEGREE AWARDED 28 JULY 2026 University of Hyderabad, India <i>Thesis: Advances in Deep Learning for Medical Image Segmentation and Classification</i> - Advisor: Dr. Avatharam Ganivada, School of Computer and Information Sciences - Funded by the UGC Junior Research Fellowship (All India Rank 53)
Sep 2017 – Mar 2019	M.Phil. in Computer Science Mewar University, India <i>Thesis: Performance Comparison of Face Detection and Recognition Algorithms</i>
Mar 2013 – Jun 2016	Master of Computer Applications (MCA) University of Kashmir, India <i>Project: SMS Intimation System for Online Leave Management</i>
Mar 2010 – Jan 2013	B.Sc. (Mathematics, Physics, Information Technology) University of Kashmir, India

ACADEMIC APPOINTMENTS

2025 – Present	Assistant Professor School of Computer Science, UPES, Dehradun, India - Shortlisted and appointed under the Research Faculty track in recognition of research output in deep learning for medical imaging - Lead an independent research programme on interpretable deep learning for medical image analysis; author of two 2026 journal papers produced at UPES - Teach core B.Tech. and M.Tech. Computer Science courses; design course material and assessments - Supervise B.Tech. and M.Tech. capstone projects in machine learning and computer vision
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Jan 2022 – Dec 2024	Teaching Assistant School of Computer and Information Sciences, University of Hyderabad, India - Delivered tutorial and laboratory sessions for graduate courses in deep learning and computer vision - Designed and evaluated programming assignments; mentored students on course projects
Jan 2021 – Dec 2022	System Administrator, Artificial Intelligence Lab School of Computer and Information Sciences, University of Hyderabad, India - Managed GPU workstations and server infrastructure supporting the AI research group - Configured and optimised deep learning software environments for lab-wide research use
2019	Assistant Professor Government Degree College Uri, Baramulla, J&K, India Taught undergraduate Computer Science courses; developed material for programming and data structures

GRANTS AND FUNDING

ICMR ANVESHAN Small Extramural Grant — Principal Investigator SUBMITTED 2026 · UNDER REVIEW

AI-Assisted Automated Quantification of Umbilical Artery Doppler Ultrasound for Early Detection of Fetal Compromise. Requested budget **₹25,00,000** over 24 months. Call No. ICMR/IDC/EPMS/CallforProposal/2026; decision expected September 2026.

- Role: Principal Investigator**, with Dr. Ufaque Muzaffar (Medical Officer, Department of Health & Family Welfare, J&K) as clinical Co-PI and Prof. Tanupriya Choudhury (UPES) as Co-Investigator
- Priority area: Reproductive, Maternal & Child Health — Antenatal Care (ICMR high-priority category)

Institution of Eminence (IoE) International Travel Grant — University of Hyderabad 2024

INR 1,00,000 covering registration, international travel and per-diem. Sanctioned 24 July 2024 under Sanction Order No. UoH-IoE/Travel/24/375, approved by the Vice-Chancellor.

- Competitively awarded to present *Dynamic Weight-Adjusted Ensemble Loss for Enhanced Medical Image Segmentation* at the 4th International Conference on Computing and Communication Networks (ICCCNet-2024), **Manchester, United Kingdom**, 17–18 October 2024

In preparation: ANRF (Anusandhan National Research Foundation) research grant proposal on uncertainty-aware deep learning for multimodal medical imaging — submission planned 2026–27.

RESEARCH PUBLICATIONS

Author name in bold. Quartile rankings per Scimago (SJR); citation counts per Google Scholar, August 2026.

PEER-REVIEWED JOURNAL ARTICLES

- M. F. Dar** and S. Mukhtar, “MSCT-Trans: Multi-Scale CNN Token Transformer for Interpretable Ultrasound Image Classification,” *Ultrasound in Medicine & Biology* (Elsevier), 2026. ACCEPTED 28 JUL 2026 · IN PRESS SCIE · Q1
First & corresponding author · independent of PhD advisor
- M. F. Dar** and A. Ganivada, “Saliency-guided AttentionNet: Dual-branch deep learning for breast ultrasound classification,” *Biomedical Signal Processing and Control*, vol. 127, art. 111194, 2026. doi: 10.1016/j.bspc.2026.111194 SCIE · Q1
- M. F. Dar** and A. Ganivada, “Fuzzy rough set loss for deep learning-based precise medical image segmentation,” *Computerized Medical Imaging and Graphics*, vol. 128, art. 102716, 2026. doi: 10.1016/j.compmedimag.2026.102716 SCIE · Q1
2 citations
- M. F. Dar** and A. Ganivada, “Adaptive ensemble loss and multi-scale attention in breast ultrasound segmentation with UMA-Net,” *Medical & Biological Engineering & Computing* (Springer), vol. 63, no. 6, pp. 1697–1713, 2025. doi: 10.1007/s11517-025-03301-5 SCIE · Q2 13 citations
- M. F. Dar** and A. Ganivada, “Deep learning and genetic algorithm-based ensemble model for feature selection and classification of breast ultrasound images,” *Image and Vision Computing*, vol. 146, art. 105018, 2024. doi: 10.1016/j.imavis.2024.105018 SCIE · Q1 36 citations
- A. N. Alhaj, N. D. Patel, A. Singh, R. K. Bondugula, **M. F. Dar**, and J. Ahamed, “Design and analysis of a robust security layer for software defined network framework,” *International Journal of Sensor Networks*, vol. 46, no. 1, pp. 1–14, 2024. doi: 10.1504/IJSNET.2024.141613 SCIE · Q3 8 citations
- M. F. Dar** and A. Ganivada, “EfficientU-Net: A novel deep learning method for breast tumor segmentation and classification in ultrasound images,” *Neural Processing Letters*, vol. 55, no. 8, pp. 10439–10462, 2023. doi: 10.1007/s11063-023-11333-x SCIE · Q2 44 citations · most cited

- [8] S. Mukhtar, **M. F. Dar**, and A. Kaur, "Latent fingerprint enhancement and matching using intuitionistic type-2 fuzzy," *International Journal of Artificial Intelligence and Soft Computing*, vol. 7, no. 4, pp. 313–328, 2022. doi: 10.1504/IJAISC.2022.130558 3 citations
- [9] **M. F. Dar** and S. Dixit, "Performance comparison of face detection and recognition algorithms," *International Journal of Science and Research*, vol. 8, no. 1, pp. 986–994, 2019. doi: 10.21275/ART20194439

PEER-REVIEWED CONFERENCE PROCEEDINGS

- [10] **M. F. Dar** and A. Ganivada, "Dynamic weight-adjusted ensemble loss for enhanced medical image segmentation," in *Proc. 4th International Conference on Computing and Communication Networks (ICCCNet-2024)*, Manchester, UK. Singapore: Springer Nature, 2025, pp. 1–12. Scopus indexed

RESEARCH INTERESTS AND SCHOLARLY SERVICE

RESEARCH INTERESTS

- Deep learning architectures for medical image analysis and computer-aided diagnosis
- Loss function design: fuzzy rough set, adaptive and dynamically weighted ensemble objectives
- Interpretable and explainable AI: saliency-guided and attention-based model design
- Medical image segmentation, classification and cross-dataset generalisation
- Soft computing, fuzzy-rough learning and evolutionary feature selection

PEER REVIEW – JOURNALS

- IEEE Transactions on Medical Imaging
- IEEE Journal of Biomedical and Health Informatics
- Information Fusion (Elsevier)
- Image and Vision Computing (Elsevier)
- Engineering Applications of Artificial Intelligence (Elsevier)
- Neural Computing and Applications (Springer)
- Multimedia Tools and Applications (Springer)
- Journal of Clinical Ultrasound (Wiley)

EDITORIAL SERVICE & OPEN SCIENCE

- Editorial Board Member, *PriMera Scientific Engineering* (ISSN 2834-2550)
- Research code released open source, including *Saliency-Guided AttentionNet* (github.com/MohsinFurkh)

HONOURS AND AWARDS

- 2024 **IoE International Travel Grant**, University of Hyderabad – INR 1,00,000 for international conference presentation in Manchester, UK
- 2019 **UGC NET with Junior Research Fellowship (JRF)**, Computer Science & Applications – **All India Rank 53**

PRESENTATIONS AND PROFESSIONAL DEVELOPMENT

- Oct 2024 **Conference Presenter – ICCCN-2024, Manchester, United Kingdom**
"Dynamic Weight-Adjusted Ensemble Loss for Enhanced Medical Image Segmentation" – funded by the IoE International Travel Grant
- 2026 One-week Train-the-Trainer (T3) programme, *Cloud Application Development*, **IBM**
- 2025 One-week Train-the-Trainer (T3) programme, *Pattern Recognition*, **IBM**
- 2024 Workshop participant, *MRI and EEG Data Analysis*, IIIT Hyderabad
- 2021 Workshop participant, *Hands-On Natural Language Processing*, Machine Learning India (MLI)

TEACHING

COURSES TAUGHT (ASSISTANT PROFESSOR)

- Introduction to Programming using C
- Python Programming
- Data Structures and Algorithms
- Database Management Systems

COURSES ASSISTED (TEACHING ASSISTANT, UNIVERSITY OF HYDERABAD)

- Deep Learning for Computer Vision
- Advanced Machine Learning
- Neural Networks and Applications
- Research Methodology

COURSES PREPARED TO TEACH

- **Undergraduate:** Programming (C / Python), Data Structures and Algorithms, Database Management Systems, Operating Systems fundamentals, Introduction to Artificial Intelligence, Digital Image Processing
- **Postgraduate:** Machine Learning, Deep Learning, Computer Vision, Medical Image Analysis, Soft Computing and Fuzzy Systems, Explainable AI, Research Methodology

SUPERVISION, MENTORSHIP AND APPROACH

- Mentored 10+ IMTech, B.Tech. and M.Tech. students on deep learning and computer vision projects

- Guided two research assistants on fuzzy rough learning and mineral prospectivity classification
- Co-authored peer-reviewed work with a postgraduate student (MSCT-Trans, Ultrasound in Medicine & Biology, 2026)
- Project-based instruction pairing mathematical foundations with reproducible implementation, using active research problems to bring current literature into the classroom

INSTITUTIONAL SERVICE AND ADMINISTRATION

2025 – Present	Curriculum delivery and student project supervision, School of Computer Science, UPES
2021 – 2022	System Administrator, Artificial Intelligence Lab, School of Computer and Information Sciences, University of Hyderabad

TECHNICAL SKILLS

Programming	Python, C, MATLAB, SQL
Deep Learning	TensorFlow, Keras, PyTorch (working knowledge), CUDA / GPU training workflows
Machine Learning	Scikit-learn, NumPy, Pandas, SciPy, OpenCV, Matplotlib, Seaborn
Tools	Git, Docker, Jupyter, PyCharm, LaTeX, Linux / HPC, Mendeley, Zotero
Databases	MySQL
Domain	Medical image segmentation and classification, explainable AI, fuzzy and rough set methods, evolutionary optimisation

REFERENCES

Dr. Avatharam Ganivada

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